Model Evaluation

Intermediate

Three closely related metrics that get confused constantly — and when to reach for each.

MAE

$$MAE = (1/n)\Sigma|y_i - \hat{y}_i|$$

The average absolute size of the error, in the original units of y. Easy to explain to a non-technical stakeholder.

MSE

$$MSE = (1/n)\Sigma(y_i - \hat{y}_i)^2$$

Same idea as MAE, but squared — so large errors count disproportionately more.

RMSE

$$RMSE = \sqrt{MSE}$$

Takes MSE back into the original units of y, while still punishing large errors more heavily than MAE does.

Metric	Easy to interpret?	Penalizes large errors?	Units
MAE	Yes	Less	Same as target
MSE	No	Strongly	Squared
RMSE	Yes	Strongly	Same as target

INTERVIEW TIP

"Why use RMSE instead of MAE?" — a strong answer: RMSE is more sensitive to large, costly mistakes (useful when big errors are especially bad, like structural engineering estimates), while MAE treats every unit of error the same (useful when you want a metric that isn't dominated by a few outliers).

R^2 Score

Intermediate

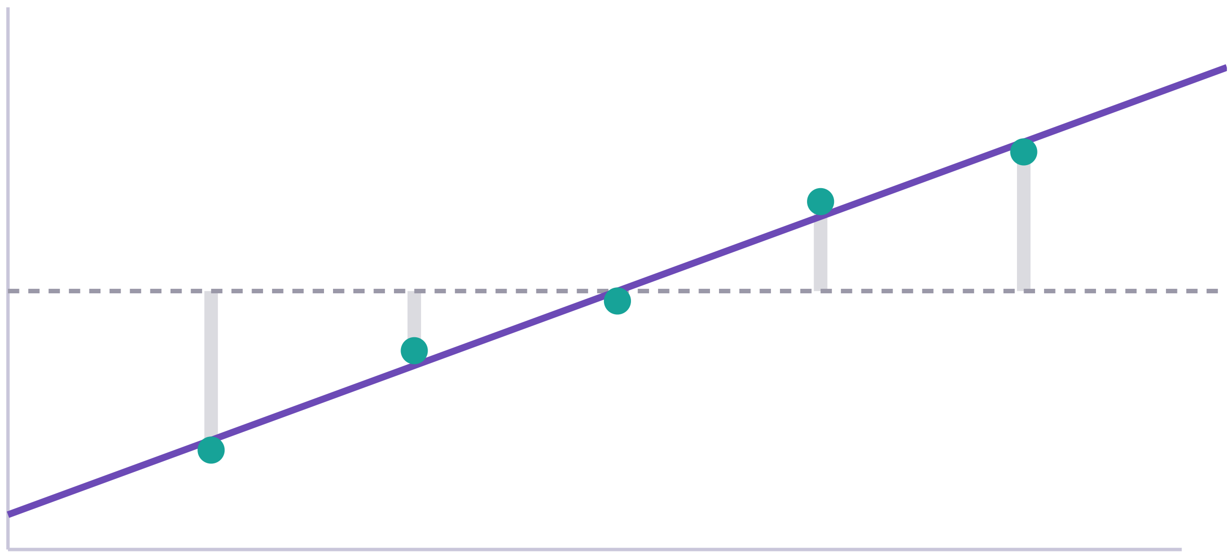
The most quoted, most misunderstood metric in regression.

FORMULA

$$R^2 = 1 - SSE/SST$$

`SSE` is your model's total squared error. `SST` ("total sum of squares") is the total squared error you'd get from the dumbest possible baseline model: always predicting the mean of y. R^2 asks: *how much better is my model than just guessing the average, every time?*

Interactive — R^2 Intuition



SST (grey) = 22.80 SSE (purple) = 0.30 R^2 = 0.987

Grey bars show error against the flat mean-prediction line; purple bars show error against the fitted regression line. R^2 is how much smaller the purple total is.

Value	Meaning
$R^2 = 1$	Perfect fit — SSE is zero
$R^2 = 0$	Your model is exactly as good as always predicting the mean
$R^2 < 0$	Your model is *worse* than just predicting the mean — it's actively making things worse (common with a badly misspecified model, or when evaluated on very different test data)

THE FALSE RULE TO UNLEARN

"Higher R^2 always means a better model" is false. R^2 mechanically increases (or at worst stays flat) every time you add *any* feature to a model, even a useless random one — which is why adjusted R^2 and out-of-sample performance matter more than raw R^2 alone.

Correlation vs. Regression

Beginner

Two closely related ideas that answer different questions.

Correlation	Regression
Symmetric: correlation of X with Y equals correlation of Y with X	Directional: defines predictor → target explicitly
A single number from −1 to 1 describing strength/direction of a linear relationship	A full equation that produces a prediction for any input

WHY STRONG CORRELATION DOESN'T PROVE CAUSATION

Ice cream sales and drowning incidents are strongly correlated — because both rise in summer heat, not because either causes the other. A third variable (temperature) drives both. Regression can quantify the association precisely without ever establishing that one variable causes the other; that requires experimental design or causal inference methods beyond this course.